

ASVA

iOS 26 Liquid Crystal UI/UX Design System

Complete visual language, colors, components, screens, animations, transitions and interaction specification

Design direction: Premium emergency operations interface using translucent liquid-crystal surfaces, soft depth, dynamic highlights and spatial motion.

Rule: Emergency clarity always overrides visual decoration.

1. Design Vision

ASVA should feel like a premium native iOS-style emergency application. The visual system uses dark atmospheric backgrounds, translucent glass layers, subtle specular highlights, soft shadows and smooth spring-like motion. The interface should feel physical rather than flat.

- Emergency information remains visually dominant.
- Glass surfaces sit above the map/background.
- Accent colors communicate state and priority.
- Motion explains state changes instead of decorating the screen.
- Every interactive element has default, pressed, loading, success and disabled states where applicable.
- Visual effects must never reduce readability of medical information.

2. Color System

Token	Value	Purpose
Deep Background	#071018	Primary app background
Primary Text	#F7FAFC	Main information
Secondary Text	#AAB6C4	Supporting information
Muted Text	#718092	Low-priority metadata
Crystal Fill	rgba(255,255,255,.10)	Primary glass
Crystal Strong	rgba(255,255,255,.16)	Elevated glass
Crystal Border	rgba(255,255,255,.22)	Glass edge
Crystal Highlight	rgba(255,255,255,.40)	Specular highlight
Critical	#FF453A	Critical emergency
High	#FF9F0A	High priority
Moderate	#FFD60A	Moderate priority
Success	#30D158	Connected / completed
Info	#0A84FF	Primary actions / route
Voice	#BF5AF2	Voice interaction

Accent colors should be used sparingly. The base visual field remains dark and translucent.

3. Liquid Crystal Material

Material	Specification	Use
Crystal 01	10% white fill, 22% white border, 20–28px blur	Cards
Crystal 02	14–16% white fill, 26% border, stronger shadow	Selected/elevated controls
Crystal 03	18–22% white fill, 30% border, stronger blur	Modal/sheets
Critical Glass	Red-tinted translucent fill + red edge	Critical case
Success Glass	Green-tinted translucent fill + green edge	Connected/completed

Use backdrop blur, saturation, a thin edge highlight and soft depth. Avoid nested glass layers beyond two levels.

4. Typography

Level	Specification	Use
Display	30–40px, semibold	Hero ETA / emergency state
Title	22–28px, semibold	Screen titles
Section	15–17px, semibold	Panel headings
Body	15–16px	Normal information
Caption	12–13px, medium	Metadata
Data	18–24px, semibold	Vitals / ETA / distance
IDs	12–13px, tabular/mono	Emergency and ambulance IDs

Use tabular numerals for vitals, ETA and distance so changing values do not cause layout shifts.

5. Spacing, Radius and Depth

Horizontal page padding: 16–20px
Card radius: 20px
Large sheet radius: 28–34px
Button radius: 17px
Icon button: 44–52px
Primary CTA height: 52–56px
Card gap: 12–16px
Section gap: 24–32px
Glass blur: 20–30px
Pointer/hover lift: 2–4px
Press scale: ~0.97

6. Buttons

Type	Appearance	Motion
Primary	Blue liquid-glass fill, white label	Lift + glow + 0.97 press scale
Critical	Red crystal fill	Press + confirmation feedback
Secondary	Neutral translucent glass	Soft highlight
Ghost	Transparent glass	Background appears on focus/press
Icon	Circular crystal	Semantic pulse/rotation where useful
Floating	Large glass pill/circle	Elevated hover state

PRIMARY CTA
Height: 54px
Radius: 17px
Fill: rgba(10,132,255,.82)
Border: 1px rgba(255,255,255,.22)
Label: 16px semibold
Pressed: scale(.97)
Success: brief green edge pulse

7. Status Language

State	Color	Motion
CRITICAL	#FF453A	1.8–2.4s restrained breathing glow
HIGH	#FF9F0A	Subtle pulse
MODERATE	#FFD60A	Static; shimmer only on update
LOW	#30D158	Static
EN ROUTE	#0A84FF	Moving location indicator

State	Color	Motion
APPROACHING	#BF5AF2	Soft ring pulse
ARRIVED	#30D158	One-time confirmation pulse
OFFLINE	#718092	Static / reconnect animation only

8. Global Navigation

Keep the emergency context persistent. On mobile, use full-screen views with floating glass controls and bottom sheets. On desktop, use a map as the primary canvas and glass side panels for operational information.

Mobile:

MAP / CONTENT

↓

Floating controls

↓

Bottom glass sheet

Desktop:

Queue		
/ details		LIVE MAP

9. Paramedic: Emergency Start

Top: ASVA + connection state

Main: Registered ambulance ID

Hero: READY FOR EMERGENCY

Primary: [START EMERGENCY]

Secondary: history / settings

The registered Ambulance ID is preloaded. Patient name remains optional and must not block the start flow.

10. Paramedic: Active Emergency

ASVA ● LIVE

AMB-1047

EMG-82931 CRITICAL

PATIENT RECEIVED

[FIND HOSPITAL]

Live status / connection / location

Emergency ID and severity should be immediately scannable.

11. Hospital Selection

Present glass hospital cards containing distance, beds, emergency readiness and ETA. The recommended card receives a brighter edge and shallow elevation.

Hospital A

4.2 km 12 beds ETA 09 min

Emergency ready

[SELECT HOSPITAL]

On selection, the card expands subtly, the route preview fades in and the navigation sheet morphs into place.

12. Navigation / Live Map

Top glass pill:
Hospital • ETA • Traffic

Full-screen map

Floating glass:
Recenter • Traffic • Emergency

Bottom glass sheet:
ETA 09 min
4.2 km
Moderate traffic
[NAVIGATE]

The map is the visual anchor during transport. GPS markers should interpolate smoothly between positions.

13. Medical Monitoring

MEDICAL STATUS

HR		SpO ₂		
120 BPM		86%		

BP		Conscious.		
90/60		UNCONSC.		

[ SPEAK]
[+ REPORT]

Use large stable numbers. Critical values can have a restrained red glow and explicit text status.

14. Voice Interaction

State	Visual / motion
Idle	Crystal mic button, static
Listening	Purple/white waveform + expanding ring + breathing glow
Processing	Waveform contracts + subtle shimmer
Extracted	Structured fields appear one-by-one
Confirmation	Uncertain field gets amber edge and confirmation CTA
Error	Red edge + retry action

Voice is not merely speech-to-text. The system must extract medical entities into structured fields and require confirmation for ambiguous values.

15. Voice Extraction Animation

 Voice
↓
Crystal processing orb
↓
CONSCIOUSNESS → UNCONSCIOUS
HR → 120 BPM
SpO₂ → 86%
BP → 90/60

Field entrance: 180–250ms fade + slight upward movement. Do not animate values so aggressively that scanning becomes difficult.

16. Medical Report Upload

Upload button
↓
Glass bottom sheet
↓
ECG / Blood Test / X-Ray / CT-MRI / Prescription / Other
↓
File preview
↓
Cloudinary upload progress
↓
✓ Uploaded
↓
Metadata created

Use a circular progress indicator or glass progress bar. Upload failure must not stop the active emergency.

17. Hospital Login

Centered glass card

ASVA
Hospital Operations

[Continue with Google]

Secure hospital access

After Firebase Authentication, backend authorization must verify that the account belongs to an approved hospital.

18. Hospital Dashboard

ASVA Hospital Operations ● LIVE User ▼			
INCOMING			
□ AMB-1047 5m LIVE MAP			
□ AMB-1082 11m			
□ AMB-1031 7m □			
□ AMB-1090 4m □			

Only ambulances whose selected destination is the authenticated hospital appear on the map and queue.

19. Incoming Ambulance Card

CRITICAL ● LIVE
AMB-1047
CARDIAC
ETA 05 MIN

HR 120 BP 90/60 SpO₂ 86%

[VIEW CASE]

Queue sorting is severity first, ETA second. A selected card uses a brighter crystal edge and shallow elevation.

20. Hospital Case Detail

AMB-1047 • EMG-82931
CRITICAL • CARDIAC
ETA 05 MIN

LIVE LOCATION
CURRENT VITALS
TREATMENT
MEDICAL REPORTS

On desktop, use a glass inspector/side panel over the map. On mobile, use a full-screen sheet.

21. Report Viewer

Top: Report type + timestamp
Center: original document/image
Bottom: Close / authorized action
Background: blurred case context

Opening a report triggers backend authorization before controlled access to the Cloudinary asset.

22. Traffic Control Interface

Traffic control should be visually simpler than the medical dashboard. Show ambulance, location, route, destination, ETA and priority. Do not expose medical reports or unnecessary patient details.

23. Motion System

Motion	Duration	Use
Micro	120-180ms	Button press / toggle
Component	220-320ms	Cards / panel entrance
Sheet	350-450ms	Bottom sheet / modal
Marker	300-700ms	Interpolated GPS movement
Status	300-500ms	Severity / ETA changes
Major transition	500-700ms	Screen-level transition
Critical pulse	1.8-2.4s	Critical status only

24. Transition Rules

- Hospital selection → navigation: preserve map continuity; morph the glass panel.
- Navigation → medical monitoring: spatial transition while retaining emergency ID.
- Incoming ambulance: animate into its sorted queue position.
- Severity increase: brief critical edge pulse, then settle.
- ETA update: crossfade/number transition rather than abrupt replacement.
- Report upload success: progress ring morphs into checkmark.
- Modal close: reverse the opening motion.

25. Liquid Crystal Physics

- Glass highlights shift subtly with pointer/focus position.
- Controls compress approximately 3% on press.
- Floating elements may rise 2-4px on hover.
- Sheets settle with spring-like motion.
- Parallax should be subtle and disabled when reduced motion is requested.
- Avoid continuous animation on non-critical UI.

26. Loading States

Glass skeleton: moving low-contrast shimmer
 Map: dark atmospheric placeholder
 Voice: crystal orb + waveform
 Upload: circular progress
 Hospital list: translucent skeleton rows
 Dashboard reconnect: small live-status indicator

27. Error and Empty States

Condition	UI behavior
No suitable hospital	Glass warning + retry + concise explanation
GPS unavailable	Location unavailable banner + retry; never show stale data as current
Voice failure	Retry voice + manual input
Ambiguous medical field	Highlight field + require confirmation
Report upload failure	Retry upload; emergency continues
Offline	Persistent connection state + safe synchronization/retry

28. Accessibility

- Maintain readable contrast over translucent surfaces.
- Never communicate severity by color alone; include text and/or icon.
- Touch targets minimum 44×44px.
- Support reduced-motion preferences.
- Do not hide essential information behind animation.
- Provide semantic labels for assistive technologies.
- Use stable large numerical typography for vitals and ETA.

29. Responsive Layout

Device	Behavior
Mobile	Single column, full-screen map, bottom glass sheets, thumb-friendly controls
Tablet	Map + medical panel / queue split
Laptop	Hospital map + incoming queue + case inspector
Large desktop	Persistent queue + large map + detail inspector

30. Design Tokens

```
--bg: #071018
--text-primary: #F7FAFC
--text-secondary: #AAB6C4
--critical: #FF453A
--high: #FF9F0A
--moderate: #FFD60A
--success: #30D158
--info: #0A84FF
--voice: #BF5AF2

--glass-fill: rgba(255,255,255,.10)
--glass-fill-strong: rgba(255,255,255,.16)
--glass-border: rgba(255,255,255,.22)
--glass-highlight: rgba(255,255,255,.40)

--radius-card: 20px
--radius-sheet: 30px
```


--radius-button: 17px
--blur-glass: 24px
--touch-target: 44px

31. Component Inventory

Component	States
GlassButton	Default / Hover / Pressed / Disabled / Loading / Success
StatusPill	Critical / High / Moderate / Low / Live / Offline
VitalCard	Normal / Warning / Critical / Updating
AmbulanceCard	Default / Selected / Critical / Arrived
HospitalCard	Available / Recommended / Selected / Unavailable
VoiceButton	Idle / Listening / Processing / Success / Error
UploadCard	Idle / Uploading / Success / Failed
MapMarker	Moving / Selected / Approaching / Arrived
GlassSheet	Collapsed / Expanded / Closing
Toast	Info / Success / Warning / Error

32. Microcopy

- Use short action labels: Find Hospital, Select Hospital, Start Navigation, Speak, Confirm, View Report.
- Avoid long explanatory text during active emergencies.
- Use exact labels such as HR, BP, SpO₂ and ETA.
- Critical alerts state what changed and what action is required.
- Avoid ambiguous labels such as Proceed when a specific action can be named.

33. Complete User Journey

START
↓
Registered ambulance
↓
START EMERGENCY
↓
Hospital recommendations
↓
SELECT HOSPITAL
↓
Route + traffic + ETA
↓
Live location
↓
Voice medical input
↓
Structured vitals + confirmation
↓
Upload report
↓
Hospital notified
↓
Hospital priority queue
↓
Case details
↓
Cloudinary report viewer
↓

ARRIVED
↓
COMPLETED

34. Visual QA Checklist

- Glass is translucent but readable.
- Dark atmospheric background remains consistent.
- Critical state is instantly recognizable.
- Primary CTA is obvious at every step.
- Animations are short and purposeful.
- Map is the visual anchor during navigation.
- Hospital queue is scannable within seconds.
- Vitals and ETA use stable large numerals.
- No medical information is obscured by effects.
- Mobile controls are thumb-friendly.
- Desktop dashboard balances information density with hierarchy.

35. Final UI/UX Direction

ASVA should look like a premium emergency operating system rather than a conventional healthcare dashboard. Liquid-crystal materials create hierarchy through depth, translucency, light and motion. Emergency severity, ETA, location and medical values always take priority over visual effects.

Design mantra: Clear under pressure. Fluid in motion. Precise in information.